

# TrES-1b

R-1780

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_TRES-1\_190905 · created 2026-08-02T23:29:10+00:00

## BENCHMARK: INACCURATE

### Results

|                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| Mid-Transit Time (Tmid)                           | 2458731.83867 +/- 0.00064 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.1845 +/- 0.0037                 |
| Transit depth (Rp/Rs)^2                           | 3.4 +/- 0.14 [%]                  |
| Orbital Inclination (inc)                         | 89.92 +/- 0.52                    |
| Airmass coefficient 1 (a1)                        | 1.663 +/- 0.021                   |
| Airmass coefficient 2 (a2)                        | -0.1692 +/- 0.0014                |
| Scatter in the residuals of the lightcurve fit is | 1.13 %                            |
| Best Comparison Star                              | None                              |
| Optimal Aperture                                  | 4.73                              |
| Optimal Annulus                                   | 22.04                             |
| Transit Duration (day)                            | 0.10844 +/- 0.00062               |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                             |
|---------------------------|---------------------------------------------|
| Rp/R■ fit vs published    | 0.1845 ± 0.0037 vs 0.1358 (deviation 9.43σ) |
| Duration fit vs published | 0.10844 d vs 0.1042 d (deviation 4.94σ)     |
| Mid-time O–C              | 5.3 min                                     |
| Depth SNR                 | 35.7                                        |
| Residual scatter          | 1.13 %                                      |

### Light curve

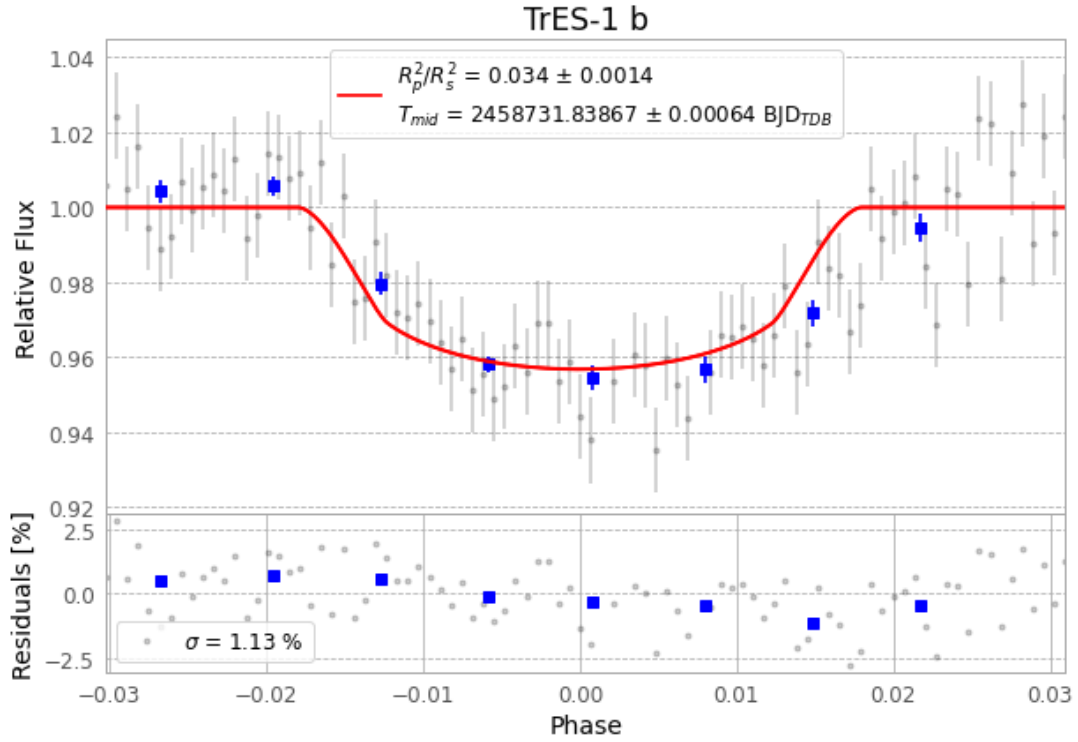


Figure 1 — FinalLightCurve\_TrES-1 b\_2019-09-04.png (SHA-256 2aa683ed101b2208...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [278, 244], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 b9d9e2fd3e0a1346...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

90 FITS frames (59 MB), TRES-1190905055113.FITS → TRES-1190905101812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-190905.log (403b735a6824...), FinalLightCurve\_TrES-1 b\_2019-09-04.png (2aa683ed101b...), FinalLightCurve\_TrES-1 b\_2019-09-04.csv (6f5a8aec187f...)

## Compute resources

Wall time 195.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 23:29Z.